

Research on SiC MOSFET Model Based on Simulink

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Abstract—Based on the Simulink tool within the Matlab platform and taking SiC MOSFET power devices as a reference, an equivalent circuit model of nonlinear capacitors C_{ds} and C_{gd} was established, and a SiC MOSFET model was constructed. In addition, its dynamic and static characteristics were analyzed according to the model. The simulation results show that the output characteristics at different temperatures match the original curve well; The rise and fall times of dynamic characteristics are consistent with the comparison errors in the data manual, which verifies the correctness of the model and has certain feasibility.

Keywords—Matlab/Simulink; MOSFET; Curve fitting

I. INTRODUCTION

With the growing market demand for power device functionality, MOSFET devices made from Silicon Carbide (SiC) material have been developing rapidly. These devices feature higher voltage tolerance, faster switching speeds, and better thermal resistance [1], thus successfully reducing the cost and losses in power device production. However, as research progresses, more precise device models are needed, which include both static and dynamic models. When studying dynamic models, it is crucial to consider their nonlinear characteristics, particularly the gate-drain capacitance (C_{gd}) and drain-source capacitance (C_{ds}). The impact of parasitic capacitance on dynamic characteristics is substantial and directly affects model accuracy. Therefore, researching dynamic models is crucial for establishing precise circuit models [2] [3] [4].

Although PSpice software is relatively mature in SiC MOSFET modeling, reference [5] provides a accurate LTSPICE model based on an improved EKV MOSFET model, its limited capability in handling mathematical equations often results in lower accuracy of the constructed device models. In contrast, Matlab/Simulink software has become an ideal choice due to its powerful simulation and data processing capabilities. Reference [6] using Simulink to establish a SiC MOSFET model configured with state-space equations. Reference [7] studied a commercial power SiC MOSFET model in Simulink. Compared to traditional SiC MOSFET modeling methods, Matlab/Simulink enables the establishment of device models from a perspective that considers multiple physical effects.

This paper conducts modeling research on the equivalent circuit of the parasitic capacitance C_{ds} and C_{gd} using

Matlab/Simulink, and construct SiC MOSFET model. We conducted simulations of the model, and the results showed good consistency with the data from the datasheet, thereby validating the accuracy of the model. This method not only ensures the rigor of the model, but also contributes to improving its precision. Thus, promoting the application and optimization of SiC MOSFET devices.

II. DYNAMIC AND STATIC MODELS OF SiC MOSFET

A. Static model of SiC MOSFET

The static characteristics of SiC MOSFET mainly refer to its static transfer characteristics and static output characteristics, describing the relationship between drain current (I_d), gate source voltage (V_{gs}), and drain source voltage (V_{ds}). Due to their extremely short dynamic switching times, SiC MOSFET enter saturation. In this region, drain current and voltage differences increase significantly, exerting a significant impact on the overall losses of the MOSFET. Key equations in the static model include the following:

When $V_{gs} \leq V_{th}$:

$$I_d = 0 \quad (1)$$

Equation (1) represents the drain current I_d when the device is in cutoff and not conducting.

When $V_{gs} > V_{th}$ and $V_{gs} - V_{th} > V_{ds}$, the current I_d expression is as follows:

$$I_d = \frac{W}{L} K_p \left[(V_{gs} - V_{th}) V_{ds} - \frac{1 + \alpha}{2} V_{ds}^2 \right] (1 + \lambda V_{ds}) \quad (2)$$

Equation (1.2) is the expression for the drain current in the linear region, where K_p is the intrinsic transconductance parameter of the device, W/L is the channel width-to-length ratio, V_{th} is the threshold voltage, λ is the channel length modulation parameter, and α represents the slope of threshold voltage variation with temperature.

When $V_{gs} > V_{th}$ and $-V_{th} > V_{ds}$, the expression for the current I_d in the saturation zone is as follows:

$$I_d = \frac{WK_p}{2L(1 + \alpha)} (V_{gs} - V_{th})^2 (1 + \lambda V_{ds}) \quad (3)$$

B. Dynamic model of SiC MOSFET

The dynamic characteristics of SiC MOSFETs are

primarily influenced by their switching behavior, with parasitic capacitance being a crucial factor. When SiC MOSFETs are used in circuits with inductive loads, parasitic capacitance in the circuit play a critical role in the switching characteristics. In the circuit, there are three parasitic capacitance: the input capacitance C_{iss} , the output capacitance C_{oss} , and the reverse transfer capacitance C_{rss} [8][9]. Figure 1 shows the dynamic equivalent circuit model of a SiC MOSFET.

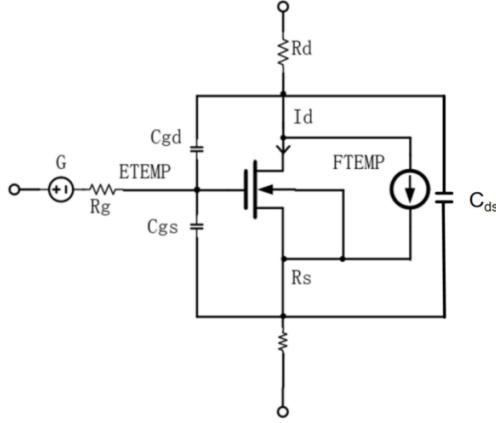


Fig. 1. Dynamic Equivalent Circuit Model of SiC MOSFET

According to the above diagram, there is a junction capacitance in the circuit, which mainly affects the circuit during charging and discharging when turned on and off. The main focus of this article on the dynamic modeling of devices is to construct a junction capacitance model. The relationship between junction capacitance and capacitance value can be obtained from the relationship between input capacitance C_{iss} , output capacitance C_{oss} , and reverse transmission capacitance C_{rss} given in the data manual:

$$\begin{aligned} C_{ds} &= C_{oss} - C_{rss} \\ C_{gs} &= C_{iss} - C_{rss} \\ C_{gd} &= C_{rss} \end{aligned} \quad (4)$$

Among them, C_{gs} is the gate source capacitance, C_{gd} is the gate drain capacitance, and C_{ds} is the drain source capacitance.

III. SIMULINK MODEL OF SICMOSFET

A. Constructing a capacitor C_{ds} model in SIMULINK

Extract values of V_{ds} , C_{ds} , and C_{gd} from the datasheet [10] using tools, and use MATLAB to perform segmented curve fitting to obtain the relationship between V_{ds} and C_{ds}/C_{gd} :

$$C_{ds} = \begin{cases} 4906 \cdot \exp(0.4542 V_{ds}) + 5286 \cdot \exp(0.0246 V_{ds}) & 0.1 V < V_{ds} < 10 V \\ 2284 \cdot \exp(0.0112 V_{ds}) + 8782 \cdot \exp(0.000466 V_{ds}) & 10 V < V_{ds} < 1000 V \end{cases} \quad (5)$$

$$C_{gd} = \begin{cases} 1988 \cdot \exp(0.6938 V_{ds}) + 3132 \cdot \exp(0.1189 V_{ds}) & 0.1 V < V_{ds} < 10 V \\ 6188 \cdot \exp(0.02055 V_{ds}) + 2503 \cdot \exp(0.000589 V_{ds}) & 10 V < V_{ds} < 1000 V \end{cases} \quad (6)$$

For a capacitor, the current passing through it is $I(t)$, as shown in formula (7), where C is the capacitor. Therefore, a controlled current source can be used to describe it. The nonlinear capacitance C_{gd} can be described by the product of

a fixed capacitance of $C=1F$ and a function $f(v_{ds})$, thus obtaining the equivalent model shown in Figure 2 and Figure 3.

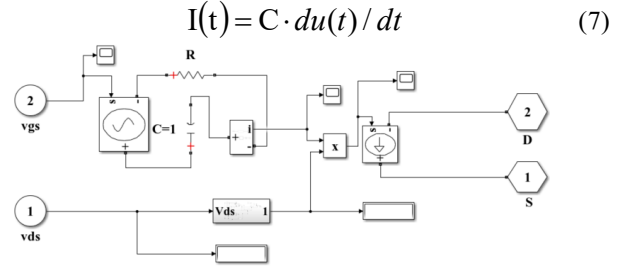


Fig. 2. C_{ds} Capacitance Model

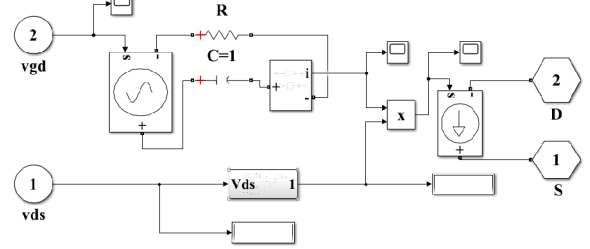


Fig. 3. C_{gd} Capacitance Model

In the upper part of the diagram, the capacitor module consists mainly of a resistor, a fixed capacitor, and a voltage source. The lower part is primarily composed of expressions (5) and (6), which together form the C_{ds} and C_{gd} capacitance model.

B. SiC MOSFET Model

On the basis of fitting calculations in Matlab/Simulink, a SiC MOSFET model was successfully established. As shown in Figures 4, the model takes into account the parasitic resistance of SiC MOSFETs, including R_d , R_g , and R_s , representing different internal device resistances. These resistors are set to model values of 0.04Ω , 6.8Ω , and 0.05Ω , respectively.

Based on fitting calculations in Matlab/Simulink, the SiC MOSFET model has been successfully established, as shown in Figure 4.

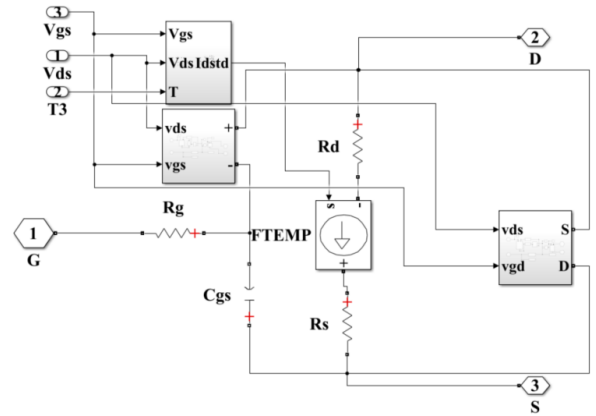


Fig. 4. SIMULINK Model of SiC MOSFET

The model includes a temperature compensation voltage source ETEMP for calculating drain current, a compensation current source module FTEMP, and a capacitor module. Additionally, there are two internal parasitic capacitance: C_{gd} and C_{gs} . Since the capacitance C_{gs} changes very little with the bias voltage V_{ds} , it is considered as a fixed capacitance in the model. Simultaneously, parasitic resistances of the SiC MOSFET are also considered, including R_d , R_g , and R_s . Their respective values are $0.04\ \Omega$, $6.8\ \Omega$, and $0.05\ \Omega$.

IV. SIMULATION AND ANALYSIS OF SiC MOSFET MODEL

A. Static Characteristics Simulation of SiC MOSFET

Figure 5 depicts the output characteristics of the SiC MOSFET at 25°C , corresponding to gate voltages from 10V to 16V. At lower V_{ds} , the relative error is small, and the simulation results closely match the experimental data. However, as V_{ds} increases into the saturation region, the error also increases. Particularly under larger V_{gs} conditions, the gap between simulation and experimental data significantly widens. Overall, there is good consistency between experimental and simulated results, validating the model's output characteristics. As shown in Figure 6, increasing the simulation temperature and V_{ds} voltage results in a smaller gap in the linear region, aligning more closely with the device's behavior.

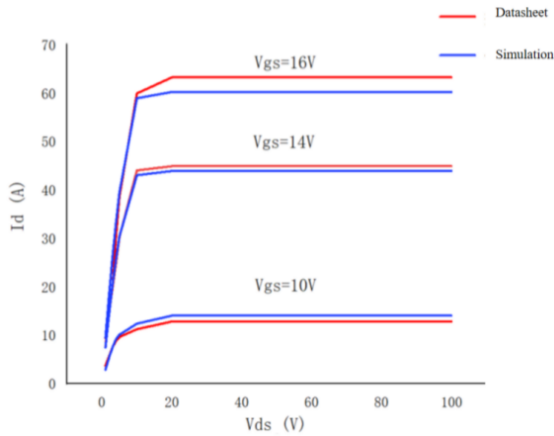


Fig. 5. Output Characteristic Curves of Simulation and Actual Device at 25°C

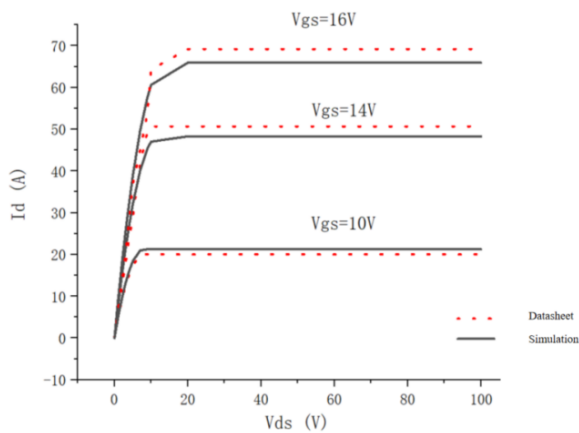


Fig. 6. Output Characteristic Curves of Simulation and Actual Device at 150°C

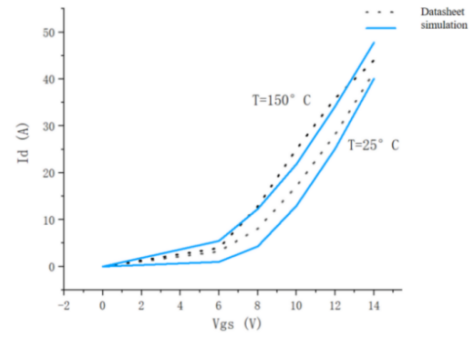


Fig. 7. Transfer Characteristic Curves at Different Temperatures

According to Figure 7, the simulation curves exhibit a trend consistent with the datasheet, albeit with some numerical deviations. At a particular temperature, when the gate-to-source voltage (V_{gs}) is low, the transconductance gradually increases with increasing V_{gs} . Furthermore, as the temperature rises, the rate of increase in transconductance also accelerates. These observations validate the accuracy of the model.

B. Simulation of dynamic characteristics of SiC MOSFET model

As shown in Figure 8, this is the test circuit used in the device manual for constructing MOSFET switching circuits. According to the guidance in the device manual, the test circuit shown in Figure 9 was designed to evaluate the switching characteristics of the SiC MOSFET model. As depicted in the figure, the parasitic capacitance C_{ds} model has been added onto the existing framework and encapsulated.

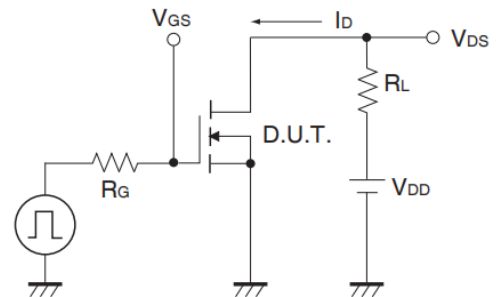


Fig. 8. Test Circuit from the Device Manual

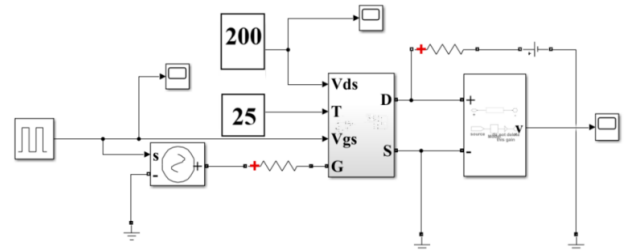


Fig. 9. SIMULINK Test Circuit

In this model, we set the amplitude of the pulse voltage signal to 10V, the duty cycle to 50%, the period to 0.00001

seconds, and the drain source voltage V_{ds} to 200V. Meanwhile, resistors R1 and R2 are $5K\Omega$. The results are shown in Figure10. The figures below respectively show the V_{ds} wave forms of the modeled device during its on and off states.

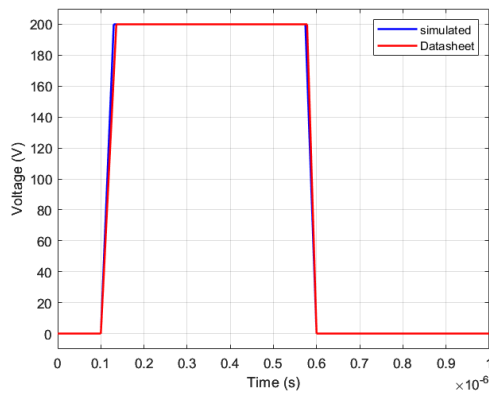


Fig. 10. Waveform of model on and off

From the simulation results in the above figure, The simulated curve roughly matches the curve in the data manual, it can be seen that the measured rise time is 30.62ns and the fall time is 26.12ns. The turn-on time obtained from the device manual is 36ns and the turn-off time is 22ns. The two times are close, which verifies the correctness of the modeling.

V. CONCLUSION

This paper employed curve fitting methods to study the relationship between capacitance C_{ds} , C_{gd} and V_{ds} . By designing subcircuits that simulate the parasitic capacitance variation with voltage, an equivalent model for parasitic capacitance was established. Leveraging Simulink tools and building upon traditional SiC MOSFET modeling, a more accurate SiC MOSFET model was developed. Using datasheet information, test circuits were constructed to analyze the device's dynamic and static characteristics. Simulation results were compared with actual device data, confirming the model's accuracy within permissible error margins. This study is of great significance in the following aspects:

- By accurately simulating the parasitic capacitance model, the accuracy of the SiC MOSFET model has been improved, making up for the shortcomings of traditional models in describing capacitance changes.
- Combining Simulink tools for simulation to make the model more accurate in predicting dynamic and static characteristics.

- The methods and results of this study provide valuable references for modeling other similar power electronic devices. Building test circuits through data manuals and conducting simulation analysis not only shortens the research and development cycle, but also reduces experimental costs, which is of positive significance for promoting the development of new technologies in the field of power electronics.

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